J Antimicrob Chemother 2025; **80**: 1302–1308
https://doi.org/10.1093/jac/dkaf070 Advance Access publication 28 March 2025

Journal of Antimicrobial Chemotherapy 50 years

Population pharmacokinetics of aciclovir and its major metabolite 9-carboxymethoxymethylguanine and safety profile of valaciclovir in early liver transplant recipients

Benjamin Kably [1]*, Mathilde Briard[2], Claire Francoz[3], Olivier Roux[3], Nadhira Houhou[4], Vincent Mackiewicz[4], Gilles Peytavin [1,5], Francois Durand[3] and Minh P. Lê [1,5,6]

[1]Pharmacology Department, APHP. Nord, Bichat-Claude Bernard Hospital, Paris, France; [2]Pharmacy, APHP. Nord, Beaujon Hospital, Clichy, France; [3]Hepatology, APHP. Nord, Beaujon Hospital, Clichy, France; [4]Virology Department, APHP. Nord, Bichat-Claude Bernard Hospital, Paris, France; [5]UMR1137, IAME, Inserm, Paris, France; [6]UFR de Pharmacie, Université Paris Cité, Paris, France

*Corresponding author. E-mail: benjamin.kably@aphp.fr

Received 18 November 2024; accepted 24 February 2025

Background: Valaciclovir is frequently prescribed for cytomegalovirus infection prophylaxis. Its major metabolite 9-carboxymethoxymethylguanine (9-CMMG), when accumulated in renally impaired patients, is neurotoxic. Its synthesis involves enzymes that could be impacted in liver transplant recipients. This retrospective study aimed to describe the pharmacokinetic (PK) and safety profile of aciclovir and 9-CMMG early after liver transplantation in patients receiving valaciclovir prophylaxis.

Methods: Consecutive (ideally five) blood samples were drawn. Plasma concentrations of aciclovir/9-CMMG were quantified by UPLC-MS/MS. Medical data were collected from digital records. A joint population PK model for aciclovir/9-CMMG was developed (Monolix 2023R1). Monte Carlo simulations were used to estimate C_{min} and AUC_{0-24}.

Results: Fifty patients (21 women) in the postoperative phase of liver transplantation were enrolled, with median age of 56.0 years and median weight of 69.5 kg; 255 samples were collected 19.0 days after transplantation. No drug–drug interaction was reported. A one-compartment model with first-order absorption best described the pharmacokinetics (PK). Covariate analysis showed that aciclovir and 9-CMMG clearances correlated with estimated glomerular filtration rate (eGFR). In normorenal patients, receiving valaciclovir 2000 mg q8h, estimated AUC_{0-24} values were 44.8 and 13.3 mg·h/L for aciclovir and 9-CMMG, respectively. The median estimated metabolic ratio of AUC_{0-24} (9-CMMG/aciclovir) was 30.4% and 129.9% for patients with >90 and <30 mL/min/1.73 m^2 eGFR, respectively. There were no valaciclovir-related adverse events during hospitalization.

Conclusions: This model allowed the PK and basal metabolic ratio of aciclovir and 9-CMMG in early liver transplantation to be defined. The correlation with renal function suggests important implications for therapeutic drug monitoring of these compounds, which will need confirmation in different cohorts.

Introduction

Cytomegalovirus (CMV) is one of the leading causes of opportunistic infection in solid organ transplantation, especially in the early stages.[1,2] Valaciclovir is one of the most commonly prescribed drugs for CMV infection prophylaxis in liver transplantation, as it has been proven to decrease the incidence of CMV infection post-orthoptic liver transplantation from 23%–85% to <10%, with satisfying cost-effectiveness.[3] The pharmacokinetics (PK) of aciclovir involve hepatic metabolism leading to several metabolites such as 9-carboxymethoxymethylguanine (9-CMMG), its main metabolite, and other minor metabolites like 8-hydroxy-aciclovir.[4] It has been previously described that 9-CMMG, the main metabolite of aciclovir, may very probably be responsible for its neurotoxic adverse events, especially related to its accumulation when using high-dose regimens and/or in renally impaired patients.[5–7] Details about the clinical expression of these types of episodes have been described as

© The Author(s) 2025. Published by Oxford University Press on behalf of British Society for Antimicrobial Chemotherapy. All rights reserved. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.

Downloaded from academic.oup.com/jac/article/80/5/1302/8099162 by PIERRE MARQUET user on 21 October 2026

aciclovir-induced neuropsychiatric symptoms[8] or linked with the so-called Cotard's syndrome.[9] The synthesis pathways of 9-CMMG remain unclear, but it has been suggested that they involve liver enzymes such as alcohol dehydrogenase and aldehyde dehydrogenase.[10] The metabolism of aciclovir is likely to be affected by the progressive recovery of liver functions during the early post-transplant course. However, in the long term, liver transplant recipients may also undergo pathophysiological changes that can alter drug PK to an extent that is currently not well known.[11–13] For all these reasons, together with the long-term duration of the treatment, therapeutic drug monitoring (TDM) of aciclovir and 9-CMMG could be useful for these patients. In this particular setting, aciclovir treatment is, most of the time, initiated orally via its prodrug valaciclovir. To the best of our knowledge, the PK features of aciclovir and 9-CMMG in liver transplant patients has never been extensively described. Furthermore, data are lacking regarding safety and efficacy in these patients. The objective of this study was to describe the PK and safety profiles of both aciclovir and 9-CMMG in early phase liver transplant patients receiving valaciclovir for CMV infection prophylaxis.

Materials and methods

Patients and data collection

A retrospective monocentric study was conducted in hospitalized adult liver transplant recipients regardless of the indication for transplantation. Patients included those who received oral prophylaxis with valaciclovir and benefited from aciclovir and 9-CMMG TDM in early stages of post-liver transplantation. Routine aciclovir and 9-CMMG TDM consisted of 3–5 consecutive blood samples (ideally 0, 1, 2, 3.5 and 5 h post-dose), for the calculation of the 24 h AUC (AUC_{0-24}). A database was constructed including the aciclovir and 9-CMMG plasma concentrations, and the clinical, biological and therapeutic features of the included patients, taken from their electronic medical record (ORBIS® v08052100.11020.FR). The estimated glomerular filtration rate (eGFR) values for each patient were calculated with the CKD-EPI formula,[14] indexed to body surface area (BSA), considering the absence of height data for several patients of our cohort. For the safety analysis, according to the literature, a trough plasma concentration (C_{min}) toxicity threshold of 4.0 mg/L[15] was used for aciclovir, and 2.6 mg/L for 9-CMMG.[5] Medical records were reviewed for any clinical signs of neurotoxicity (headaches, dizziness, confusion, drowsiness, hallucinations, tremors, seizures) or other potential aciclovir-related adverse events (acute kidney injury, allergic reactions), coinciding with valaciclovir treatment.

Plasma drug determination

Blood samples were immediately centrifuged and the decanted plasma was stored at −20°C until analysis. On the day of the analysis, plasma concentrations of aciclovir and 9-CMMG were simultaneously quantified by UPLC coupled with tandem MS (Xevo UPLC-TQD, Waters®, Milford, MA, USA), using a slightly modified method.[16] The method was linear from the lower limit of quantification of 0.025 mg/L up to 10 mg/L for both compounds.

CMV serology

The pre-transplant CMV serostatus was determined from serum samples using a chemiluminescence immunoassay on a Liaison XL analyser (Liaison CMV IgG II kit, DiaSorin, Italy). Signal of CMV IgG antibodies above 14 UA/mL was considered positive, otherwise negative.

Statistical analysis

The results are presented as either percentages for qualitative variables or median and IQR for quantitative variables.

Population PK (popPK) analysis

A non-linear mixed-effects model approach was chosen to study both aciclovir and 9-CMMG plasma concentrations using the Monolix® suite (version 2023R1, Lixoft, France) in a joint popPK model. First, the estimation of the population parameters without the inclusion of any covariates was evaluated using the SAEM algorithm and the estimation of the standard error using a stochastic approach. Additive, proportional and combined models were used to evaluate residual variability. The performance of candidate models was assessed using both diagnostic goodness-of-fit (GOF) plots and statistical criteria (Bayesian information criteria; BIC) to select among competing structural models. The impact of potential correlations between continuous covariates (age, weight, eGFR calculated with the CKD-EPI formula, time since transplantation) and categorical covariates (sex, geographical origin, renal replacement therapy) was assessed. A stepwise forward inclusion approach was employed to identify significant covariates. Each covariate was incorporated univariately into the parameter of interest and was selected based on a reduction in the objective function value (OFV) by a minimum of 3.84 units (which corresponds to $P<0.05$ as determined by the χ^2 test), a decrease in inter-individual variability, and enhancements in model fit. The covariate exhibiting the most substantial alteration in the OFV was retained within the model, and this procedure was reiterated until no covariates maintained statistical significance. GOF plots include population predicted (PRED) versus observed aciclovir and 9-CMMG plots, and individual predicted (IPRED) versus observed aciclovir and 9-CMMG plots. In addition, a prediction-corrected visual prediction check (pcVPC) was performed by simulating 500 replicates to examine the agreement between observed (aciclovir and 9-CMMG) and model-simulated data to evaluate the predictive performance of the model. The final inclusion of correlations in the structural model was determined by the precision of the parameter estimates. The clearance (CL) and volume of distribution (V) parameters of aciclovir are apparent, and expressed as CL/F and V/F, based on the bioavailability (F) of aciclovir from valaciclovir administration (F = 54%).[17]

Model-based dosing simulations

For each dosing regimen, Monte Carlo simulations with 1000 replicates were performed based on the final model. The valaciclovir regimens tested were: 1000 mg q12h, 1000 mg q8h, 1500 mg q12h, 1500 mg q8h, 2000 mg q12h and 2000 mg q8h. AUC_{0-24} values were predicted by the final model on Day 11.

Ethics

This study was conducted in accordance with the Declaration of Helsinki and the French law (Loi Jardé, 5 March 2012).

Results

Population characteristics

Fifty patients hospitalized in our centre following liver transplant surgery between June and December 2023 were included. The baseline demographic and clinical features of the patients are shown in Table 1. The population consisted of 42% (21/50) women, with a median (IQR) age and weight of 56.0 years (35.5–61.5) and 69.5 kg (60.5–84.3), respectively, and different geographical origins (66% Europe, 16% Central Africa, 12%

Downloaded from academic.oup.com/jac/article/80/5/1302/8099162 by PIERRE MARQUET user on 21 October 2026

Table 1. Baseline demographic, biological and clinical features of the population

Feature	
Number of patients (men/women)	50 (29/21)
Age (years), median (IQR)	56.0 (35.5–61.5)
Body weight (kg), median (IQR)	69.5 (60.5–84.3)
Geographical origins, *n* (%)	
Europe	33 (66)
Central Africa	8 (16)
North Africa	6 (12)
South America	2 (4)
South Asia	1 (2)
Comorbidities, *n* (%)	
Diabetes	15 (30)
Chronic kidney disease	0 (0)
Hypertension	15 (30)
Hyperlipidaemia	2 (4)
Obesity (BMI ≥ 30 kg/m^2)	8 (17.0)
Serum creatinine (μmol/L), median (IQR)	67.5 (53.4–90)
Creatinine clearance (eGFR), (mL/min/1.73 m^2) (CKD-EPI), median (IQR)	104 (83.8–122.5)
eGFR (mL/min/1.73 m^2), *n* (%)	
>90	33 (66)
61–90	13 (26)
31–60	3 (6)
<30	1 (2)
AST (U/L), median (IQR)	26 (19–37)
ALT (U/L), median (IQR)	55 (26–94)
GGT (U/L), median (IQR)	136 (76–191)
ALP (U/L), median (IQR)	130 (99–169)
Bilirubin (total, μmol/L), median (IQR)	11 (7–20)
Bilirubin (conjugated, μmol/L), median (IQR)	6 (4–15)
Positive CMV serology, *n* (%)	41 (82)
Number of samples/patient, median (IQR)	5.0 (5.0–5.0)
Sampling day after liver transplantation (days), median (IQR)	20.0 (15.3–23.8)
Indication for liver transplantation, *n* (%)	
Alcoholic liver disease	13 (26)
Autoimmune hepatitis	9 (18)
HBV	5 (10)
HCV	5 (10)
Acute liver failure	4 (8)
Non-alcoholic steatohepatitis	4 (8)
Mixed	4 (8)
Cholangiocarcinoma	1 (2)
Others	5 (10)
Valaciclovir dose/day (g/day), median (IQR)	4.0 (4.0–6.0)
Valaciclovir dosing regimen, *n* (%)	
2000 mg q8h	17 (34)
2000 mg q12h	24 (48)
1500 mg q8h	6 (12)
1500 mg q12h	2 (4)
1000 mg q12h	1 (2)

ALP, alkaline phosphatase.

North Africa, 4% South America, and 2% South Asia). The indications for liver transplantation were alcohol-related liver disease in 26%, viral hepatitis in 20%, autoimmune hepatitis in 18%, acute liver failure in 8%, metabolic dysfunction-associated steatohepatitis in 8%, and miscellaneous causes in 20%. All patients were treated for CMV prophylaxis, and received oral valaciclovir at dosing regimens ranging from 1 to 6 g/day, depending on their renal function, body weight and the CMV virological status of both donor and recipient. Patients had various comorbidities such as diabetes (30%), hypertension (30%) and dyslipidaemia (4%). The majority of patients had a normal renal function, with 3 (6%) and 1 (2%) patients having an eGFR (CKD-EPI) of 31–60 and <30 mL/min/1.73 m^2, respectively. No patient in our cohort had renal replacement therapy. Before surgery, virological data showed that 41 (82%) patients had positive CMV serology. Nine patients who had a negative CMV IgG detection before transplantation had a CMV PCR performed. Among them, no seroconversion was detected. In addition, no CMV-related diseases were noted in the early follow-up of our entire cohort. Regarding aciclovir and 9-CMMG TDM, a total of 255 samples were collected, a median (IQR) of 19.0 (15.0–24.0) days after surgery, with the large majority of patients having five consecutive samples drawn (one and three patients had three and four samples, respectively). There were two patients (4%) with an aciclovir C_{min} of >4 mg/L, of whom 1 (2%) had a 9-CMMG C_{min} of >2.6 mg/L. All patients were treated with tacrolimus and mycophenolate mofetil for transplant rejection prophylaxis. No drug–drug interaction was detected, and there were no aciclovir-related adverse events during the hospitalization.

PopPK model

A one-compartment model with first-order absorption best described the PK. Typical relative standard errors [RSE(%)] shown in Table 2 were: lag time (T_{lag}) 0.3 h (fixed), absorption rate constant (k_a) 1.2 h^{-1} (20.7%), apparent volume of distribution (*V*/F) 752 L (7.7%), apparent clearance (CL/F) 19.9 L/h (32.4%) for aciclovir and 234.7 L/h (13.3%) for 9-CMMG with a tissue partition coefficient (K_p) 0.1 h^{-1} (10.4%). A proportional error model was used for aciclovir and a combination of an additional and proportional error model was used for 9-CMMG. Covariate analysis showed that aciclovir and 9-CMMG CL/F values were significantly correlated with eGFR ($P < 0.01$), with regression coefficients of β CL/F [log(eGFR)] 2.9 (20.5%) for aciclovir and β CL [log(eGFR)] 0.6 (36.2%). Inter-individual variability was estimated for all parameters. All shrinkage values for aciclovir and 9-CMMG were <20%.

In patients with normal eGFR receiving 2000 mg q8h, the estimated median (IQR) AUC_{0-24} values were 44.8 mg·h/L (30.5–64.9) and 13.3 mg·h/L (7.7–22.9) for aciclovir and 9-CMMG, respectively. The median metabolic ratio of AUC_{0-24} (9-CMMG/aciclovir) was 30% and 116% in patients with >90 and <30 mL/min/1.73 m^2 eGFR, respectively.

Model diagnostics

GOF plots for both aciclovir and 9-CMMG (PRED versus observed, iPRED versus observed, pcVPC) are shown in Figures 1 and 2,

Downloaded from academic.oup.com/jac/article/80/5/1302/8099162 by PIERRE MARQUET user on 21 October 2026

Table 2. PopPK parameters of aciclovir and 9-CMMG from the final model

Parameter	Value	RSE (%)	Coefficient of variation (%)
Aciclovir			
T_{lag} (h)	0.3	—	
k_a (h^{-1})	1.19	20.7	
V/F (L)	751.19	7.72	
CL/F (L/h)	19.87	32.4	
β CL/F	2.86	20.5	
9-CMMG			
CL (L/h)	234.71	13.3	
β CL	0.62	36.2	
K_p (h^{-1})	0.1	10.4	
Standard deviation of the random effects			
ω_{ka}	1.02		22.1
ω_V	0.39		14.7
$\omega_{CL(aciclovir)}$	0.87		33.2
$\omega_{CL(9\text{-}CMMG)}$	0.79		13.5
ω_{Kp}	0.58		16.6
Correlations			
Corr K_p(9-CMMG) CL(9-CMMG)	0.55	27.8	
Error model parameters			
b (aciclovir plasma concentration, mg/L)	0.35	5.16	
a (9-CMMG plasma concentration, mg/L)	0.035	31.8	
b	0.24	9.82	

β, regression coefficient of the covariate analysis.

respectively, and in Figures S1 to S5 (available as Supplementary data at *JAC* Online). All GOF plots indicate good predictive performance of the model.

Monte Carlo simulations

The median (IQR) AUC_{0-24} and C_{min} of aciclovir and 9-CMMG, according to the eGFR and after 10 days of treatment, are presented in Tables 3 and 4, respectively. The median (IQR) metabolic ratio of AUC_{0-24} (9-CMMG/aciclovir) were 30.4% (16.9%–48.4%) and 129.9% (87.1%–164.0%) for patients with eGFR >90 and <30 mL/min/1.73 m^2, respectively (Table 5).

Discussion

This work has provided a detailed description of the combined PK profiles of aciclovir and its main metabolite 9-CMMG, as well as the expected metabolic ratio of these compounds, in the early post-operative period. The safety of oral valaciclovir therapy in our patients was good, mainly because the majority of them did not have impaired kidney function. Thus, no aciclovir-related adverse events were recorded in this study, and very few C_{min} values exceeded the toxic thresholds for aciclovir and 9-CMMG with the dosing regimens used (mostly 2000 mg q12h and 2000 mg q8h). This finding appears consistent with established knowledge regarding the neurotoxicity of aciclovir, for which the imputability is likely attributed to an accumulation of 9-CMMG in the CNS of patients with impaired kidney function,[8] due to the blockade of its primary elimination pathway.[18,19] In this regard, our study confirmed the role of valaciclovir in the management strategy of liver transplant patients. Other strategies, such as the use of valganciclovir, have been

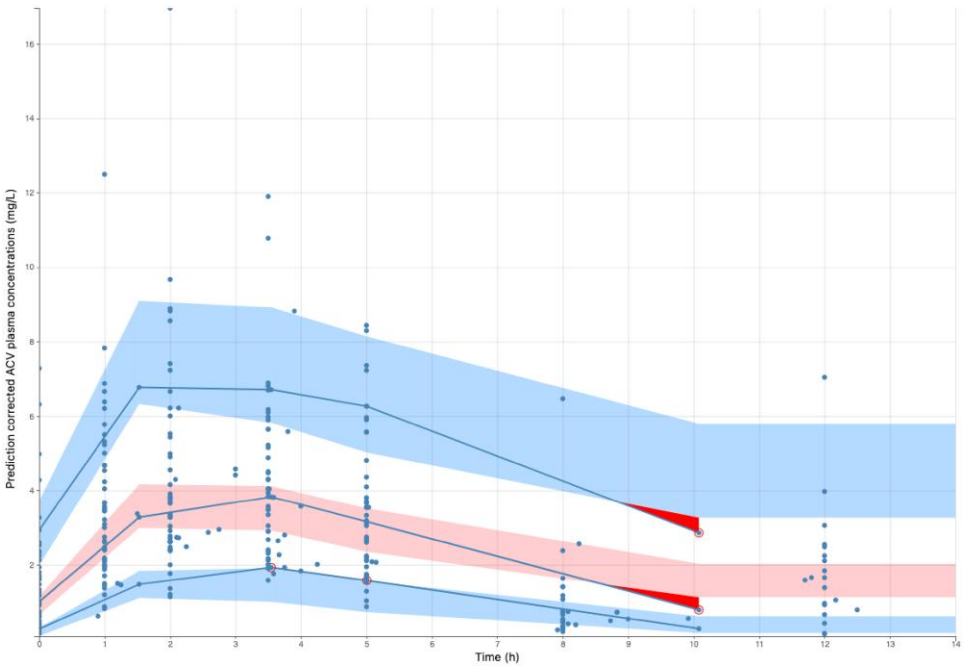


Figure 1. GOF plot: visual predictive check plot for prediction-corrected aciclovir (ACV) plasma concentrations (mg/L) over time (h).

Downloaded from academic.oup.com/jac/article/80/5/1302/8099162 by PIERRE MARQUET user on 21 October 2026

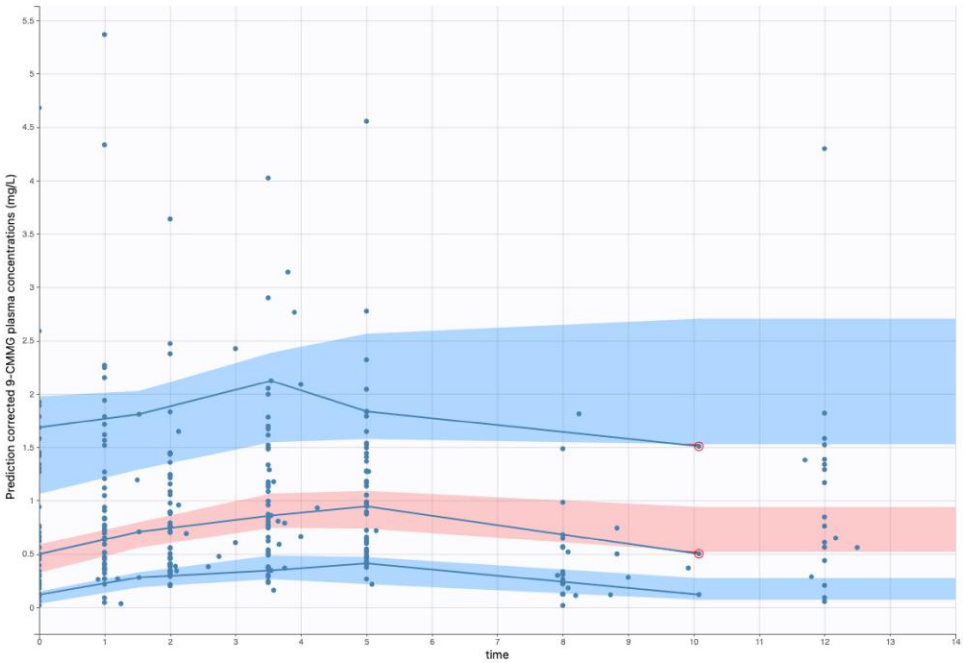


Figure 2. Goodness-of-fit (GOF) plot: pcVPC for 9-CMMG plasma concentrations (mg/L) over time (h).

Table 3. Median (IQR) Monte Carlo simulated AUC_{0-24} of aciclovir and its metabolite 9-CMMG according to the eGFR (mL/min/1.73 m^2)

	Valaciclovir dosing regimen					
	1000 mg q12h	1000 mg q8h	1500 mg q12h	1500 mg q8h	2000 mg q12h	2000 mg q8h
eGFR			Aciclovir			
>90 (*n*=684)	14.9 (10.2–21.6)	22.4 (15.3–32.5)	22.4 (15.2–32.5)	33.6 (22.9–48.7)	29.9 (20.3–43.4)	44.8 (30.5–64.9)
61–90 (*n*=236)	21.6 (15.9–29.6)	32.3 (23.9–44.3)	32.3 (23.9–44.3)	48.5 (35.8–66.5)	43.1 (31.8–66.5)	64.7 (47.4–88.7)
31–60 (*n*=60)	21.1 (14.4–45.7)	31.7 (21.5–68.5)	31.7 (21.5–68.5)	47.5 (32.3–102.8)	42.4 (28.7–91.4)	63.6 (43.1–137.1)
<30 (*n*=20)	29.6 (18.1–56.3)	44.3 (27.1–84.4)	44.3 (27.1–84.4)	66.5 (40.6–126.6)	59.1 (36.1–112.5)	88.7 (54.2–168.7)
eGFR			9-CMMG			
>90 (*n*=684)	4.4 (2.6–7.6)	6.7 (3.9–11.5)	6.7 (3.9–11.5)	10.0 (5.8–17.2)	8.9 (5.1–15.3)	13.3 (7.7–22.9)
61–90 (*n*=236)	7.7 (4.4–13.5)	11.6 (6.6–20.2)	11.6 (6.6–20.2)	17.4 (9.9–30.3)	15.5 (8.8–27.0)	23.2 (13.1–40.4)
31–60 (*n*=60)	12.6 (7.5–18.9)	18.9 (11.2–28.4)	18.9 (11.2–28.4)	28.3 (16.8–42.6)	25.2 (15.0–37.8)	37.7 (22.5–56.7)
<30 (*n*=20)	34.4 (29.5–47.4)	51.6 (44.3–71.1)	51.6 (44.3–71.1)	77.4 (66.4–106.7)	68.8 (59.0–94.8)	103.1 (88.6–142.2)

There were 1000 simulated patients for each dosing regimen.

recommended,[20] with little to no difference in terms of adverse events and acute rejections,[21] but may have inferior cost-effectiveness.[3]

To our knowledge, this is the first developed joint popPK model of aciclovir and 9-CMMG, in a cohort of liver transplant recipients in the early post-operative period. The popPK model was then used to conduct Monte Carlo simulations to predict the respective plasma AUC_{0-24} and C_{min} of aciclovir and 9-CMMG following oral administration of valaciclovir. Indirectly, this led to the conclusion that eGFR was the only variable statistically correlated with the AUC_{0-24} of these two compounds, and thus with the plasma exposure of aciclovir and 9-CMMG. Consequently, the median (IQR) metabolic ratio of AUC_{0-24} (9-CMMG/aciclovir) was calculated at 30.4% (16.9%–48.4%) in patients with normal renal function, and could reach up to 129.9% (87.1%–164.0%) in patients with an eGFR of <30 mL/min/1.73 m^2. There was no impact of the daily dose on the metabolic ratio. This finding suggests a potential benefit of the combined TDM of aciclovir and 9-CMMG, particularly for patients with impaired kidney function, as stated by previous work on the subject,[5] which will need to be confirmed in different cohorts (patients with renal insufficiency and/or experiencing adverse events). It was also noted that the exposure to 9-CMMG was much more importantly influenced by a patient's renal function than the exposure to aciclovir. In fact, compared with patients with a normal renal function, patients with an eGFR of <30 mL/min/1.73 m^2 showed a 2.0-fold increase in

Downloaded from academic.oup.com/jac/article/80/5/1302/8099162 by PIERRE MARQUET user on 21 October 2026

Table 4. Median (IQR) Monte Carlo simulated C_{min} of aciclovir and its metabolite 9-CMMG according to the eGFR (mL/min/1.73 m^2)

	Valaciclovir dosing regimen					
	1000 mg q12h	1000 mg q8h	1500 mg q12h	1500 mg q8h	2000 mg q12h	2000 mg q8h
eGFR			Aciclovir			
>90 (*n*=684)	0.2 (0.1–0.5)	0.5 (0.2–1.0)	0.4 (0.1–0.7)	0.8 (0.4–1.3)	0.4 (0.2–0.9)	1.0 (0.5–1.9)
61–90 (*n*=236)	0.4 (0.2–0.7)	0.9 (0.6–1.5)	0.6 (0.3–1.1)	1.3 (0.7–2.2)	0.9 (0.5–1.6)	1.7 (1.0–2.8)
31–60 (*n*=60)	0.4 (0.2–1.2)	0.9 (0.4–2.5)	0.6 (0.3–1.8)	1.5 (0.8–3.2)	0.9 (0.4–2.0)	1.8 (0.8–4.2)
<30 (*n*=20)	0.7 (0.4–1.3)	1.2 (0.7–2.0)	1.3 (0.8–2.0)	1.7 (0.9–3.1)	2.0 (0.8–3.7)	3.7 (1.8–6.1)
eGFR			9-CMMG			
>90 (*n*=684)	0.1 (0.05–0.3)	0.2 (0.1–0.5)	0.2 (0.1–0.4)	0.3 (0.2–0.6)	0.2 (0.1–1.0)	0.5 (0.2–0.9)
61–90 (*n*=236)	0.2 (0.1–0.4)	0.4 (0.2–0.8)	0.4 (0.2–0.7)	0.7 (0.3–1.2)	0.5 (0.2–1.0)	0.8 (0.5–1.6)
31–60 (*n*=60)	0.4 (0.2–0.8)	0.8 (0.4–1.2)	0.6 (0.3–1.2)	1.1 (0.6–2.0)	0.9 (0.3–1.5)	1.3 (0.6–2.3)
<30 (*n*=20)	1.3 (1.1–2.2)	2.1 (1.4–3.5)	2.2 (1.4–2.8)	**3.0 (2.2–3.3)**	**2.3 (2.0–3.3)**	**3.8 (3.2–5.9)**

Bold values represent trough plasma concentrations above the toxicity threshold (4 mg/L for aciclovir and 2.6 mg/L for 9-CMMG. There were 1000 simulated patients for each dosing regimen.

Table 5. Median (IQR) Monte Carlo simulated AUC_{0-24} metabolic ratio of 9-CMMG/aciclovir according to the eGFR

eGFR (mL/min/1.73 m^2) (CKD-EPI)	AUC_{0-24} metabolic ratio (9-CMMG/aciclovir) (%)
>90	30.4 (16.9–48.4)
61–90	34.9 (21.0–63.1)
31–60	47.6 (31.6–99.8)
<30	129.9 (87.1–164.0)

aciclovir AUC_{0-24} compared with a 7.7-fold increase in 9-CMMG AUC_{0-24}, as shown in Table 3.

The limitations of our study primarily lie in the small sample sizes within the lower eGFR categories and among patients with extreme body weights. For greater accuracy, the model will need validation in these categories of patients. Additionally, the good tolerance of valaciclovir in our cohort did not allow for testing the PK/pharmacodynamic (PK/PD) relationship of 9-CMMG exposure and its toxicity. The model, as well as the understanding of aciclovir metabolism, would have benefited from urinary concentrations of aciclovir and 9-CMMG, as the fraction of aciclovir metabolized to CMMG could be quantified, as well as the volume of distribution of the metabolite. We also chose not to include minor metabolites to the model, such as 8-hydroxy-aciclovir, that present with very low plasma concentrations, but could maybe have a potential impact on valaciclovir tolerability.[19]

The *V*/F and CL/F values for aciclovir found in our model appear to be higher than those reported in the literature. This could be explained by the physiological changes related to liver transplantation. First, a decrease in serum albumin levels may increase the free fraction available for distribution into the different compartments. Additionally, an increase in the extracellular volumes (oedema, ascites) in these patients may lead to a higher *V* for hydrophilic or weakly lipophilic compounds, which is the case for aciclovir. Finally, hepatic perfusion may have increased due to graft vascularization. The CL of 9-CMMG was also found to be high, which could likely be related to the low K_p value, especially since a correlation between these two parameters was found.

Finally, the long-term efficacy of the valaciclovir prophylactic treatment remains to be evaluated, especially since the simulated C_{min} levels of aciclovir often appear to be below 1.0 mg/L in normorenal patients.

In summary, we were able to provide a joint popPK model for both aciclovir and 9-CMMG in the context of CMV prophylaxis, in uncomplicated liver transplant recipients. The developed model allowed definition of the PK and basal metabolic ratio of aciclovir and 9-CMMG in this clinical setting. The significant correlation between renal function and the plasma exposure to aciclovir and 9-CMMG underlines the importance of the combined TDM of these compounds. This will need to be confirmed in different cohorts (patients with renal insufficiency, experiencing adverse events).

Funding

This study was conducted as part of our routine work.

Transparency declarations

Minh P. Lê has received travel grants from ViiV Healthcare. Gilles Peytavin has received travel grants, consultancy fees and honoraria from Gilead Sciences, ViiV Healthcare, Merck and Takeda.

Supplementary data

Figures S1–S5 are available as Supplementary data at *JAC* Online.

References

1 Razonable RR, Emery VC; 11th Annual Meeting of the IHMF (International Herpes Management Forum). Management of CMV infection and disease in transplant patients. 27–29 February 2004. *Herpes* 2004; **11**: 77–86.

Downloaded from academic.oup.com/jac/article/80/5/1302/8099162 by PIERRE MARQUET user on 21 October 2026

2 Meesing A, Razonable RR. New developments in the management of cytomegalovirus infection after transplantation. *Drugs* 2018; **78**: 1085–103. https://doi.org/10.1007/s40265-018-0943-1

3 Leong RWL, Smith DW, Garas G *et al*. Aciclovir or ganciclovir universal prophylaxis of cytomegalovirus infection in liver transplantation: an economic analysis. *Intern Med J* 2004; **34**: 410–5. https://doi.org/10.1111/j.1445-5994.2004.00567.x

4 Soul-Lawton J, Seaber E, On N *et al*. Absolute bioavailability and metabolic disposition of valaciclovir, the L-valyl ester of acyclovir, following oral administration to humans. *Antimicrob Agents Chemother* 1995; **39**: 2759–64. https://doi.org/10.1128/AAC.39.12.2759

5 Helldén A, Odar-Cederlöf I, Diener P *et al*. High serum concentrations of the acyclovir main metabolite 9-carboxymethoxymethylguanine in renal failure patients with acyclovir-related neuropsychiatric side effects: an observational study. *Nephrol Dial Transplant* 2003; **18**: 1135–41. https://doi.org/10.1093/ndt/gfg119

6 Gentry JL, Peterson C. Death delusions and myoclonus: acyclovir toxicity. *Am J Med* 2015; **128**: 692–4. https://doi.org/10.1016/j.amjmed.2015.03.001

7 Helldén A, Lycke J, Vander T *et al*. The aciclovir metabolite CMMG is detectable in the CSF of subjects with neuropsychiatric symptoms during aciclovir and valaciclovir treatment. *J Antimicrob Chemother* 2006; **57**: 945–9. https://doi.org/10.1093/jac/dkl067

8 Lindström J, Helldén A, Lycke J *et al*. An unexpectedly high occurrence of aciclovir-induced neuropsychiatric symptoms in patients treated for herpesvirus CNS infection: a prospective observational study. *J Antimicrob Chemother* 2019; **74**: 3565–72. https://doi.org/10.1093/jac/dkz357

9 Helldén A, Odar-Cederlöf I, Larsson K *et al*. Death delusion. *BMJ* 2007; **335**: 1305. https://doi.org/10.1136/bmj.39408.393137.BE

10 de Miranda P, Good SS. Species differences in the metabolism and disposition of antiviral nucleoside analogues: 1. Acyclovir. *Antivir Chem Chemother* 1992; **3**: 1–8. https://doi.org/10.1177/095632029200300101

11 Li M, Zhao Y, Humar A *et al*. Pharmacokinetics of drugs in adult living donor liver transplant patients: regulatory factors and observations based on studies in animals and humans. *Expert Opin Drug Metab Toxicol* 2016; **12**: 231–43. https://doi.org/10.1517/17425255.2016.1139575

12 Morales Junior R, Telles JP, Kwiatkowski SY-C *et al*. Pharmacokinetic and pharmacodynamic considerations of antibiotics and antifungals in liver transplantation recipients. *Liver Transpl* 2023; **29**: 91–102. https://doi.org/10.1002/lt.26517

13 Shimojima N, Shimazu M, Kikuchi H *et al*. Serum alcohol dehydrogenase: a sensitive biomarker of ongoing graft function after liver transplantation. *Clin Transplant* 2007; **21**: 498–501. https://doi.org/10.1111/j.1399-0012.2007.00676.x

14 Levey AS, Stevens LA, Schmid CH *et al*. A new equation to estimate glomerular filtration rate. *Ann Intern Med* 2009; **150**: 604–12. https://doi.org/10.7326/0003-4819-150-9-200905050-00006

15 Abdul-Aziz MH, Alffenaar J-WC, Bassetti M *et al*. Antimicrobial therapeutic drug monitoring in critically ill adult patients: a position paper. *Intensive Care Med* 2020; **46**: 1127–53. https://doi.org/10.1007/s00134-020-06050-1

16 Ärlemalm A, Helldén A, Karlsson L *et al*. Rapid determination of acyclovir, its main metabolite 9-carboxymethoxymethylguanine, ganciclovir, and penciclovir in human serum using LC-MS/MS. *Biomed Chromatogr* 2022; **36**: e5315. https://doi.org/10.1002/bmc.5315

17 Base de Données Publique des Médicaments. Résumé des caractéristiques du produit—VALACICLOVIR BIOGARAN 500 mg, comprimé pelliculé. https://base-donnees-publique.medicaments.gouv.fr/affichageDoc.php?specid=62357468&typedoc=R.

18 Unita R, Adachi K, Inada A *et al*. A decreasing plasma concentration of a toxicologically active metabolite 9-carboxymethoxymethylguanine after dialysis—a potential new clinical biomarker for improving encephalopathy in patients treated with acyclovir. *J Infect Chemother* 2023; **29**: 1068–70. https://doi.org/10.1016/j.jiac.2023.07.001

19 Smith JP, Weller S, Johnson B *et al*. Pharmacokinetics of acyclovir and its metabolites in cerebrospinal fluid and systemic circulation after administration of high-dose valacyclovir in subjects with normal and impaired renal function. *Antimicrob Agents Chemother* 2010; **54**: 1146–51. https://doi.org/10.1128/AAC.00729-09

20 European Association for the Study of the Liver. EASL clinical practice guidelines: liver transplantation. *J Hepatol* 2016; **64**: 433–85. https://doi.org/10.1016/j.jhep.2015.10.006

21 Vernooij RW, Michael M, Ladhani M *et al*. Antiviral medications for preventing cytomegalovirus disease in solid organ transplant recipients. *Cochrane Database Syst Rev* 2024; issue **5**: CD003774. https://doi.org/10.1002/14651858.CD003774.pub5

Downloaded from academic.oup.com/jac/article/80/5/1302/8099162 by PIERRE MARQUET user on 21 October 2026